Problem 1. (30 marks) Consider the 3×3 matrix

$$A = \begin{bmatrix} 1.8 & 0.4 & -0.4 \\ 0 & 2 & 0 \\ -0.4 & 0.8 & 1.2 \end{bmatrix}.$$

(a) (5 mark) Determine the eigenvalues of A.

(b) (10 mark) Determine the algebraic and geometric multiplicity of each eigenvalue of A.

(c) (5 marks) Explain why A is diagonalisable.

(d) (10 marks) Compute a matrix V such that $V^{-1}AV = D$, where D is a diagonal matrix. Write down both V and D.

Problem 2. (35 marks) Consider the second order linear ODE:

$$\ddot{y} + 3\dot{y} + 2y = \dot{u} + u$$

(a) (5 marks) Write down the observer canonical state-space representation $S = (A, B, C, D)$ for the above ODE.

(b) (10 marks) Suppose that $u(0) = a$. If the ODE has initial condition $y(0) = b$ and $\dot{y}(0) = c$, what is the corresponding initial condition for the state $x(t)$ in terms of a, b, c for the observer canonical form?

(c) (20 marks) Compute e^{At} for the observer canonical representation.

Problem 3. (35 points) Consider the electrical circuit shown in Fig. 1. This circuit has a capacitor with capacitance C F, an inductor with inductance L H, and a nonlinear resistive device with voltage-current characteristics given by $i_R = g(v_R)$, where g is a continuous function with the additional properties that $g(0) = 0$ and that it is strictly monotonically increasing, that is, $g(v_2) > g(v_1)$ whenever $v_2 > v_1$. This implies that g is invertible, meaning that for any i_R there is one and only one v_R such that $i_R = g(v_R)$. Therefore we can also write $v_R = g^{-1}(i_R)$. The circuit also has a constant dc voltage source v_{dc} and a controllable voltage source $v_{\mathrm{s}}(t)$. We choose as state variables for the circuit the current i_L passing through the inductor and the voltage v_C across the capacitor. The state equation for the circuit is:

$$\frac{d}{dt}\begin{bmatrix} v_C \\ i_L \end{bmatrix} = \begin{bmatrix} \frac{1}{C}(g(v_{\mathrm{s}} - v_C) - i_L) \\ \frac{1}{L}(v_C - v_{\mathrm{dc}}) \end{bmatrix}$$

Answer the following questions:

(a) (15 points) Suppose that the controllable voltage source is turned "off" to $v_s(t) = 0$ for all $t \geq 0$. Show that in this case the circuit has only one equilibrium point $(v_C^*, i_L^*)^T$ (you have to find this equilibrium point). Then using the candidate Lyapunov function

$$V(v_C, i_L) = \frac{C}{2}(v_C - v_C^*)^2 + \frac{L}{2}(i_L - i_L^*)^2$$

show that this equilibrium point is asymptotically stable.

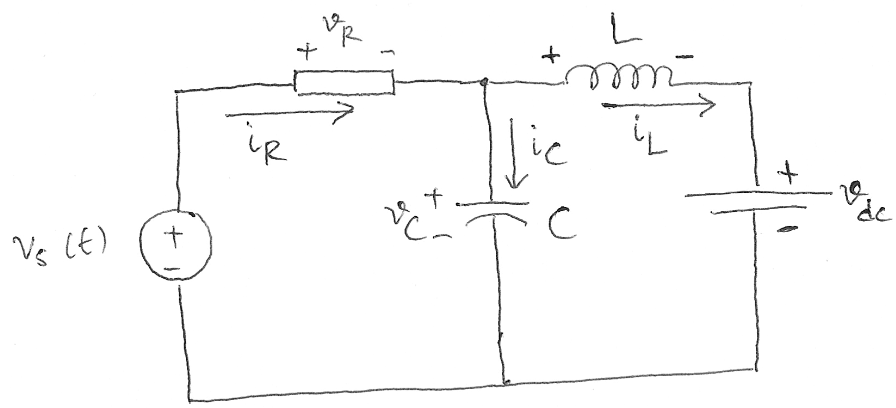

Figure 1: Electrical circuit for Problem 3.

(b) (20 points) Using the same candidate Lyapunov function as in part (a) with $v_C^* = v_{\mathrm{dc}}$, show that there is a control law for the controllable voltage source $v_s(t)$ to render any state of the form $(v_{\mathrm{dc}}, i_L^*)^\top$ asymptotically stable. Thus, $v_s(t)$ can be used to locally control the steady-state current across the inductor to any desired value i_L^*.